

Scrum-Based Process Model for Sustainable Shopping Assistant App

1. Scrum Roles

1.1 Product Owner (PO)

- Defines the vision and roadmap of the Sustainable Shopping Assistant App.
- Prioritizes the Product Backlog based on user needs, sustainability goals, and business value.
- Ensures stakeholder feedback is integrated into the development process.

1.2 Scrum Master (SM)

- Facilitates Scrum ceremonies and removes impediments.
- Coaches the team on Agile principles and Scrum practices.
- Shields the development team from external distractions.

1.3 Development Team (Dev Team)

- Cross-functional team including front-end & back-end developers, UI/UX designers, QA engineers, and data analysts.
 - Responsible for delivering potentially shippable product increments at the end of each sprint.
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2. Scrum Artifacts

2.1 Product Backlog

- A dynamic list of features, enhancements, bug fixes, and technical tasks.
- Examples:
 - User Story: "As a shopper, I want to scan barcodes to get sustainability ratings."
 - Technical Task: "Integrate barcode scanning API."

2.2 Sprint Backlog

- A subset of the Product Backlog items selected for the current sprint.
- Includes Sprint Goal, selected User Stories, and tasks needed to complete them.

2.3 Increment

- The working version of the app with added features from the Sprint.
 - Must meet the Definition of Done (DoD): code complete, tested, documented, and deployable.
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3. Scrum Ceremonies

3.1 Sprint Planning (4 hours for 2-week sprint)

- Define the Sprint Goal (e.g., "Implement barcode scanning and sustainability rating feature").
- Select backlog items and break them into tasks.

3.2 Daily Scrum (15 minutes daily)

- Quick stand-up meeting to discuss:
 - What was done yesterday?
 - What will be done today?
 - Any blockers?

3.3 Sprint Review (2 hours)

- Demonstrate the working app increment to stakeholders.
- Gather feedback for future sprints.

3.4 Sprint Retrospective (1.5 hours)

- Reflect on the sprint process:
 - What went well?
 - What could be improved?
 - Action items for the next sprint.
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4. Sample Sprint Cycle (2 Weeks)

Sprint Goal: Implement Barcode Scanning with Sustainability Ratings

Week 1:

- **Day 1 (Sprint Planning):** Select backlog items:
 - User Story 1: "As a user, I want to scan barcodes to get product sustainability ratings."
 - Task 1: Research and integrate barcode scanning library.
 - Task 2: Design UI for scanning feature.
- **Days 2-5: Development & Daily Scrums.**
 - Dev: Barcode scanner integrated.
 - Design: UI mockups finalized.
 - QA: Write test cases for barcode scanning.

Week 2:

- **Days 6-9:** Continue development.
 - Backend: Connect to sustainability data APIs.
 - QA: Begin testing barcode functionality.
- **Day 10 (Sprint Review & Retrospective):**
 - Demo barcode scanner.
 - Collect feedback.
 - Reflect on process improvements.

5. Definition of Done (DoD)

- Code reviewed and merged.
- Unit and integration tests passed.
- UI/UX meets design guidelines.
- Deployed to staging environment.
- Documentation updated.

6. Tools & Technologies

- **Project Management:** Jira or Trello
 - **Communication:** Slack, Zoom
 - **Version Control:** Git (GitHub/GitLab)
 - **CI/CD:** Jenkins, GitHub Actions
 - **Testing:** Jest, Selenium
 - **Design:** Figma, Adobe XD
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7. Backlog Refinement (Ongoing Activity)

- Regularly review and prioritize backlog items.
- Add new user stories based on stakeholder feedback.
- Break down large epics into manageable tasks.

This Scrum-based process model ensures an iterative, flexible, and user-focused approach to developing the Sustainable Shopping Assistant App, aligning with Agile principles for maximum efficiency and stakeholder satisfaction.